Simple Circuit Solver

Iteration 2 Report — System Design & Initial Implementation

Course	EECE2140
Project Leader (Iteration 2)	Brodie Newhouse
Team Members	Georgy Rached, Julia Terceros, Brodie Newhouse
GitHub Repository	https://github.com/brodienewhouse/Classic-Circuit-Calculator
Iteration Dates	Oct 18 at 12am - Oct 23 at 11:59pm
Submission Date	October 19

1. Project Leader

Project Leader for this iteration: Brodie Newhouse.
This team member has not served as Project Leader yet.

Responsibilities of the Project Leader:

- Coordinate daily progress and remove blockers
- Make sure tasks are shared fairly among all members
- Present the status report at the check-in meeting
- Keep the task table (Section 2) up to date

2. Task Definitions

Each feature of the project is listed as a separate task. Status values: Not started, Stub, In progress, Done.

ID	Task Name	Menu Option	Responsible	Files / Functions	Deliverable	Status
T1	Switch Cases/Main Menu (in terminal)	1	Georgy	tasks.cpp / task1()	Working menu loop with switch cases that call functions	Finished
T2	Component Classes	2	Brodie	tasks.cpp / task2()	Resistor and Voltage. Source classes with constructors, getters and print()	In progress
T3	Series and Parallel Resistance Functions	3	Julia	tasks.cpp / task3()	Functions that return a value based if it is series or parallel	Not Started

T4	V, I, R, P Calculator and Unit Tests	4	Julia	tasks.cpp / task4()	Ohm's law basic solver for power and every other calculation. AI-generated unit tests	Not Started
T5	Front-End and Front-End Integration	5	Georgy	tasks.cpp / task5()	User interface (built with Claude) that will be connected to the back end	Not Started
T6	Code Review and Bug Fixes	6	Julia & Brodie	tasks.cpp / task6()	Reviewed code, fixed bugs, and a test run of every menu option	Not Started

3. System Design

3.1 Classes

Each class is listed with its private data members and public methods.

Class	Private Data	Public Methods	Purpose
Resistor	name, resistance, voltage, current	getResistance(), setValues(), power(), print()	Stores a single resistor and its value
VoltageSource	name, voltage	getVoltage(), setVoltage(), print()	Represents the DC voltage source that is going to power the circuit
Circuit	vector of Resistor objects, VoltageSource, isSeries	addResistor(), equivalentResistance(), solve(), printResults()	Holds all components and does calculations for V,I,R,P

3.2 Menu Design

The program presents a text menu. Each option runs exactly one task.

Option	Menu Text	Task ID	Function Called
1	Help / How to Use	T1	task1()
2	Add Components (Source and Resistors)	T2	task2()
3	Combine Resistors (Series / Parallel)	T3	task3()
4	Solve V, I, R, P	T4	task4()
5	Launch Front-End Display	T5	task5()
6	Run Tests / Code Check	T6	task6()
0	Exit	—	—

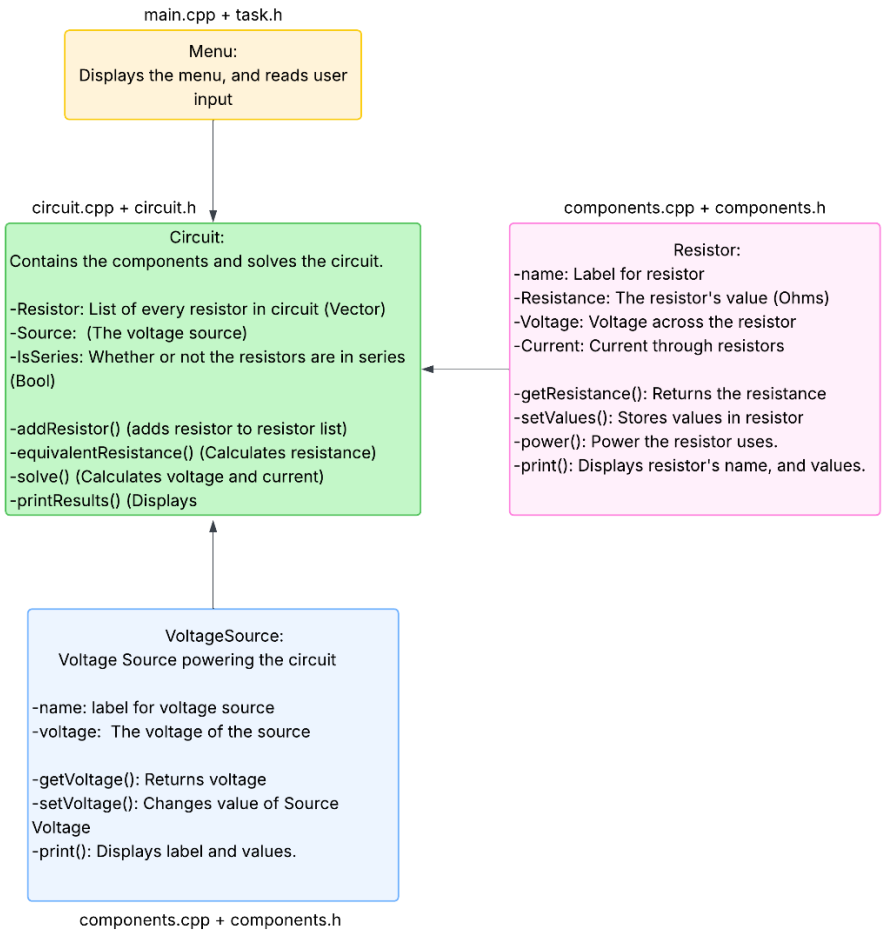
3.3 File Structure

Each class and task is assigned to a header and source file.

File	Contents	Owner
main.cpp	Main menu of program; menu loop and switch statements to choose what function to perform	Georgy
components.h / components.cpp	Resistor and VoltageSource class declarations and implementations	Brodie
circuit.h / circuit.cpp	Circuit class: series/parallel logic and V, I, R, P solver in that class	Julia
tasks.h / tasks.cpp	One function per task	All
README.md	Project description, build and run instructions	All/Assisted with AI
docs/	PDF report and Excel project plan	Project Leader

3.4 Class Diagram

The diagram below shows the three classes and how the menu code uses them.



4. Implementation

4.1 Menu and Task Dispatch

Code in Repository

4.2 Task Stubs

Code in Repository

4.3 Implementation Status

Task ID	Responsible	Status	Notes
T1	Georgy	Done	Everything works as intended
T2	Brodie	In progress	Creating components
T3	Julia	Not started (stub)	Stub in place; planned out formulas
T4	Julia	Not started (stub)	Stub in place; depends on other tasks
T5	Georgy	Not started (stub)	Stub in place; starts when backend is done
T6	ALL	Not started (stub)	Stub in place; begins when all code is done

5. Progress Documentation

Next Steps

Main documentation is in an Excel sheet.

Goals for now: finish the Resistor and VoltageSource classes (T2), write the series/parallel and V, I, R, P functions (T3, T4). Front-end will be later next month once the backend is developed fully. After that, the code will be moved onto the STM32 too.

Risks: Any delay in the backend will cause delays in frontend so it is important to get the first parts done fully and tested by early next month.